

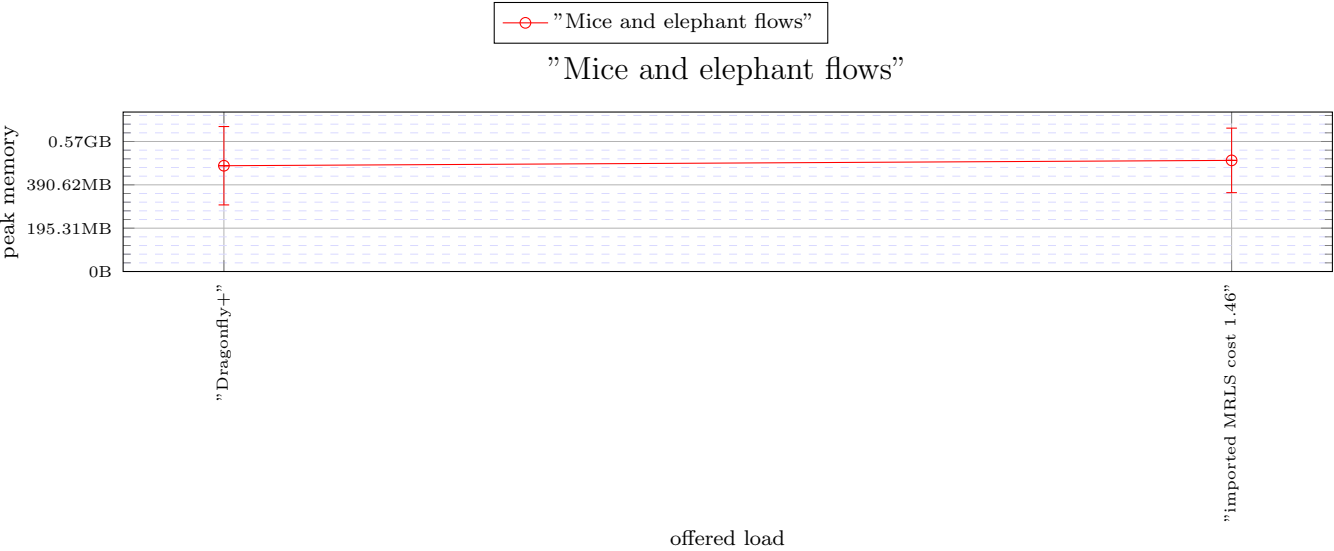


Figure 1: X: "Mice and elephant flows"

The following versions used in the simulations.

- heads/alex-dev-defe967c1db10e3b873fe074f274b6bdc9c42bb8(0.6.3)

```
Configuration{
  random_seed: ![ 42, 935, 128643 ],
  warmup: 20000,
  measured: 50000,
  statistics_packet_definitions: [
    [ ],
    [ ]],
  general_frequency_divisor: 2,
  topology: topologia![
    MultiStage{
      stages: [
        ExplicitStageFile{ filename: "../MRLS_1280_19_760_32_a_net" }],
      servers_per_leaf: 13,
      legend_name: "imported MRLS cost 1.46",
      Megafly{ global_ports_per_spine: 16, servers_per_leaf: 16, global_arrangement: Palmtree, group_size: 16, number_of_groups: 65, legend_name:
"Dragonfly+" }],
      traffic: ![
        SyntheticTrafficDistribution{
          tasks: 16640,
          load: ![ 0.1, 0.25 ],
          pattern: Uniform,
          message_size: Multimodal{
            sizes: [ 16, 256 ],
            weights: [ 0.9, 0.1 ]},
          legend_name: "Mice and elephant flows"}],
        maximum_packet_size: 16,
      router: InputOutput{
        virtual_channels: 4,
        virtual_channel_policies: topologia![
          [
            MapHop{
              hop_to_policy: [
                ArgumentVC{
                  allowed: [ 0 ]},
                ArgumentVC{
                  allowed: [ 0 ]},
                ArgumentVC{
                  allowed: [ 1 ]},
                ArgumentVC{
                  allowed: [ 1 ]},
                ArgumentVC{
                  allowed: [ 2 ]},
                ArgumentVC{
                  allowed: [ 2 ]},
                ArgumentVC{
                  allowed: [ 3 ]},
                ArgumentVC{
                  allowed: [ 3 ]}],
              VecLabel{
                label_vector: [ 0, 64, 80 ]},
              OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: true },
              OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: false },
              LowestLabel,
              EnforceFlowControl,
              Random],
            [
              MapHop{
                hop_to_policy: [
                  MapLabel{
                    label_to_policy: [
                      Chain{
                        policies: [
                          OccupancyFunction{ label_coefficient: 0, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0,
use_internal_space: true, use_neighbour_space: true, aggregate: false }],
                        Chain{
                          policies: [
                            AverageOccupancyFunction{
                              label_coefficient: 0,
                              occupancy_coefficient: 2,
                              product_coefficient: 0,
                              constant_coefficient: 16,
                              use_internal_space: true,
                              use_neighbour_space: true,
                              exclude_minimal_ports: false,
                              virtual_channels: [ 0 ]}]},
                            MapLabel{
                              label_to_policy: [
                                Chain{
                                  policies: [
                                    OccupancyFunction{ label_coefficient: 0, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0,
use_internal_space: true, use_neighbour_space: true, aggregate: false }],
                                    Chain{
                                      policies: [
                                        AverageOccupancyFunction{
                                          label_coefficient: 0,
                                          occupancy_coefficient: 3,
                                          product_coefficient: 0,
                                          constant_coefficient: 16,
                                          use_internal_space: true,
                                          use_neighbour_space: true,
                                          exclude_minimal_ports: false,
                                          virtual_channels: [ 0 ],
                                          exclude_link_classes: [ 0 ]}]}}],
                                above_policy: MapLabel{
                                  label_to_policy: [
                                    Chain{
                                      policies: [
                                        OccupancyFunction{ label_coefficient: 0, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space:
```

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true, use_neighbour_space: true, aggregate: false }]],
  Chain{
    policies: [
      OccupancyFunction{ label_coefficient: 0, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 48, use_internal_space:
true, use_neighbour_space: true, aggregate: false }]]}},
    LowestLabel,
    EnforceFlowControl,
    Random]],
  allocator: Random,
  buffer_size: 128,
  bubble: false,
  flit_size: 16,
  intransit_priority: false,
  allow_request_busy_port: true,
  output_buffer_size: 64,
  crossbar_frequency_divisor: 1,
  crossbar_delay: 2},
routing: topologia![
  Polarized{ include_labels: true, strong: false, legend_name: "Polarized" },
  MegaflyAD{
    first_allowed_virtual_channels: [ 0, 1 ],
    second_allowed_virtual_channels: [ 2, 3 ],
    minimal_to_deroute: [ 1, 1, 0 ],
    source_group_pattern: [
      [ None, None ],
      [ None, None ]],
    intermediate_group_pattern: [
      [ None, None ],
      [ None, None ]],
    destination_group_pattern: [
      [ None, None ],
      [ None, None ]],
    global_pattern_per_hop: [
      [ None, None ],
      [ None, None ]],
    legend_name: "FPAR 2 vc"}],
link_classes: [
  LinkClass{ delay: 2 },
  LinkClass{ delay: 2 },
  LinkClass{ delay: 2 }],
launch_configurations: [
  Slurm{ job_pack_size: 1, time: "3-00:00:00" }]}
```